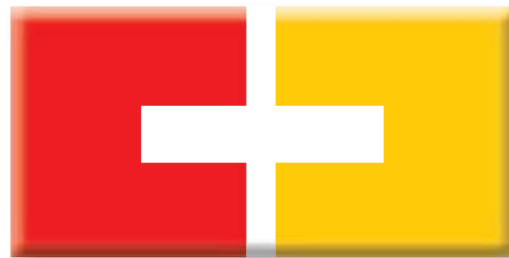


Design and Evaluation of an Adaptive FPGA Board with Modular Connector Interface

Project Work by
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ABSTRACT

This project presents the design and implementation of an adaptive FPGA-based interface board centered on the Xilinx Spartan-7 XC7S25, developed as a compact, modular, and scalable platform for embedded systems and robotics applications. The primary goal was to identify and integrate the most suitable FPGA solution that balances performance, cost, toolchain accessibility, and long-term availability. After a comparative evaluation of multiple FPGA families, the Spartan-7 was selected for its modern 28 nm architecture, low power consumption, and full support within the free Vivado WebPACK environment. The board features two 80-pin DF40C high-density connectors that enable seamless integration with external motherboards or expansion units, along with support for DDR2 memory, SPI and UART communication, and dedicated interfaces for sensors and actuators. Configuration and debugging are facilitated through onboard USB-UART and JTAG interfaces, while basic I/O interaction is supported via LEDs and push buttons. Implemented on a multilayer PCB, the design ensures efficient signal routing, stable power delivery, and domain separation for reliable operation. Tailored for educational and prototyping contexts, the board accommodates typical robotics tasks such as line-following, obstacle detection, and servo control, offering a flexible and accessible foundation for FPGA-based system development.

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Chapter 1

INTRODUCTION

In the field of embedded systems, *Field Programmable Gate Arrays (FPGAs)* play a crucial role due to their parallel processing capabilities, reconfigurability, and suitability for time-critical applications. Unlike microcontrollers, FPGAs offer hardware-level concurrency, making them ideal for robotics, signal processing, and control systems.

This project presents the design of an adaptive FPGA-based interface board built around the **Xilinx Spartan-7 XC7S25**. The board is developed as a modular platform to support a variety of sensor and actuator-driven systems. Key features include external DDR2 memory, SPI and UART communication interfaces, JTAG and USB-UART programming options, and onboard peripherals such as LEDs and push buttons. To facilitate expandability and integration, the board includes two 80-pin DF40c high-density connectors that allow reliable interfacing with external control units. A multilayer PCB layout is used to ensure clean separation of power, signal, and ground domains.

This document is organized to guide the reader through the complete development process. The initial chapters cover the motivation, design goals, and FPGA selection strategy. Subsequent sections describe the hardware features, application domains, and detailed pin mapping for both high-density connectors. The later chapters focus on peripheral integration, schematic and PCB layout overview, design evaluation, and future development directions, concluding with the tools used throughout the project.

1.1 Motivation

The motivation behind this project came from the growing demand for compact, modular hardware platforms suited for academic and prototyping environments. In university labs, developers often need to experiment with various sensors, actuators, and communication protocols. Designing a general-purpose adaptive board reduces the need to create custom hardware for each use case, saving time and increasing flexibility.

FPGAs, particularly from the Spartan-7 family, offer the reconfigurability, low power consumption, and logic capacity necessary for mid-range embedded applications. Unlike microcontrollers, they enable parallel processing and custom logic implementation, making them ideal for tasks that require real-time performance. To support clean integration with external systems, the board includes DF40c high-density connectors and is built on a multilayer PCB to ensure signal integrity.

and reduce noise. Robotics applications were a key focus, with dedicated sensors, actuators, and communication modules.

The project combines practical PCB design experience with embedded system development, resulting in a reusable FPGA-based platform tailored for academic, research, and robotics use cases.

Chapter 2

PROJECT REQUIREMENTS AND DESIGN GOALS

The goal of this project is to design a compact, low-power, and cost-effective embedded system based on the Xilinx Spartan-7 XC7S25 FPGA, tailored for educational, prototyping, and small-scale deployment use. The platform is designed to interface with a wide range of sensors, communication peripherals, and memory components, while remaining reusable and scalable.

A key requirement was selecting an FPGA with sufficient logic density, low power consumption, and broad I/O support, while ensuring compatibility with a free and accessible development toolchain. The Spartan-7 series met these criteria and supports features such as DDR2 memory, SPI, I²C, and UART communication.

To maximize modularity, the board integrates high-density DF40c connectors for expansion with external modules like motor drivers or wireless interfaces. The layout also considers future scalability by supporting pin-compatible FPGA upgrades within the Spartan-7 family.

Essential onboard peripherals including push buttons, LEDs, and oscillator circuits—support development and debugging. The use of a multilayer PCB enhances signal integrity and power distribution. All components were chosen with long-term availability in mind, making the board suitable for replication in academic and research settings.

The project's design goals focused on building a reusable and expandable embedded platform with a strong emphasis on performance, cost-efficiency, educational usability, and peripheral integration. These requirements drove every architectural and component-level decision made throughout the course of development.

Chapter 3

SELECTING THE RIGHT FPGA SOLUTION

Choosing the most suitable FPGA for a development board requires careful evaluation of multiple factors: logic capacity, I/O availability, toolchain compatibility, power consumption, cost-effectiveness, and long-term availability. These considerations become even more critical when designing an educational or prototyping-focused platform where usability, affordability, and future-proofing are essential.

In the early stages of this project, multiple FPGA families were considered, including Intel's Cyclone IV and V series, Lattice's iCE40 lineup, and Xilinx's Spartan-6 and Spartan-7 families. Each candidate offered unique advantages and trade-offs, which were evaluated across the metrics shown in Table 3.1.

Table 3.1: Comparison of FPGA Families Considered

FPGA Family	Logic Cells	Block RAM	Tool Support	Power	Open Source?
Intel Cyclone IV	22,000	1.2 Mbits	Quartus	Moderate	No
Intel Cyclone V	30,000	2.5 Mbits	Quartus Prime	Moderate-High	No
Lattice iCE40 HX8K	7,680	128 Kbits	Yosys + nextpnr	Very Low	Yes
Xilinx Spartan-6 LX25	24,051	936 Kbits	ISE (Legacy)	Moderate	No
Xilinx Spartan-7 XC7S25	23,360	1,600 Kbits	Vivado	Low	No

The **Xilinx Spartan-7 XC7S25-1FTGB196C** FPGA was selected for this project due to its strong balance of modern architecture, low power consumption, flexible packaging, and long-term scalability. Among several evaluated options—such as Intel Cyclone IV/V, Lattice iCE40, and the older Spartan-6, the Spartan-7 family stood out as a future-ready and efficient solution that is well-suited for embedded development tasks. Built on a 28 nm process node, Spartan-7 FPGAs benefit from lower dynamic power consumption and support higher operational clock frequencies, all while maintaining a compact footprint. This is particularly advantageous in thermally-constrained or battery-powered embedded systems, as the device's core voltage (VCCINT) is approximately 1.0 V, reducing overall thermal output and enabling efficient energy usage [14].

Technical Capabilities

The XC7S25 variant includes:

- 23,360 logic cells for implementing complex digital designs
- 1,600 Kbits of block RAM for buffering and temporary storage
- 80 DSP slices, allowing moderate digital signal processing operations

It supports interfacing with DDR2 SDRAM [13] and a wide range of I/O standards including LVCMOS, LVTTTL, and LVDS [14]. These features make it ideal for control logic, memory expansion, and sensor or communication interfacing.

In addition to its base architecture, the Spartan-7 series includes several advanced integrated features that enhance its utility in complex applications. The inclusion of XADC allows for on-chip voltage and temperature monitoring, which can be accessed via JTAG or user logic, aiding system diagnostics and dynamic reconfiguration. The device also incorporates MMCM and PLL clocking blocks, known as clock management tiles (CMT), which provide stable multi-domain clock generation critical for complex timing requirements. These features are all built on a unified 7-Series logic architecture, which is consistent across other Xilinx families such as Artix-7, Kintex-7, and Virtex-7, allowing for future design migration and scalability [14].

From a hardware design perspective, the FTGB196 BGA package was a strategic choice. Despite its compact size, it offers a sufficient number of I/O pins, allowing for high GPIO density while simplifying layout in multilayer PCB designs. Its fine-pitch ball grid array supports tight routing and reliable signal integrity, especially when paired with high-speed board-to-board connectors. The package's compact footprint helps maintain a small overall board size while ensuring all necessary connections for sensors, memory, and communication peripherals can be made effectively [6].

Spartan-7 also proves to be a highly suitable option for academic and prototype-oriented projects. It is fully supported by the free Vivado WebPACK toolchain [16], which includes all essential features such as IP Integrator, constraint-based synthesis, simulation, and bitstream generation. This eliminates the need for costly software licenses, making it accessible to students and researchers. Compared to more resource-intensive solutions like Intel's Cyclone V or Altera's Stratix series, Spartan-7 enables a smoother learning curve without sacrificing critical FPGA development capabilities.

Scalability and Long-Term Availability

In terms of scalability and availability, the XC7S25 model is readily stocked by major global distributors including Digi-Key and Mouser [15]. This ensures that projects relying on the chip can continue production without facing supply shortages. Additionally, the XC7S25 is often pin-compatible with larger Spartan-7 models such as the XC7S50, which enables seamless upgrades

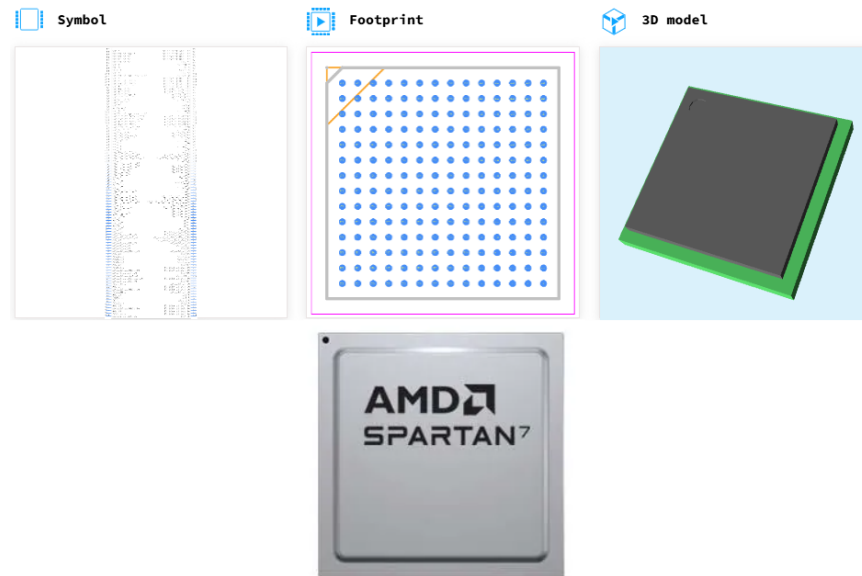


Figure 3.1: Xilinx Spartan-7 FPGA Package

in case future iterations of the project require more logic or memory resources. This upgradability reduces the need for PCB redesign and extends the long-term viability of the system architecture.

The Spartan-7 XC7S25 presents a well-rounded, accessible, and forward-compatible platform that meets the performance and flexibility demands of modern embedded system development. Its architectural features, robust toolchain support, and practical packaging make it a strong candidate for both initial prototyping and scaled deployment.

Chapter 4

FEATURES

The adaptive FPGA board was designed with a focus on modularity, flexibility, and expandability. Below are the key features that make this board suitable for embedded systems, robotics, and academic research:

- **Xilinx Spartan-7 XC7S25 FPGA:** Provides a strong balance of logic resources, low power consumption, and modern toolchain support.
- **Multilayer PCB Design:** Ensures optimized signal integrity and power distribution across functional layers.
- **Two DF40c 80-pin High-Density Connectors:** Allow secure and compact plug-and-play connection to a motherboard or expansion board.
- **Support for External DDR2 Memory:** Enables high-speed memory interfacing for real-time and data-intensive applications.
- **Multiple Communication Interfaces:** Includes dedicated lines for SPI, I2C, and UART protocols to interface with various peripherals.
- **JTAG Interface:** Allows for easy programming and debugging of the FPGA using standard tools.
- **USB-UART Bridge:** Facilitates serial communication with the FPGA for monitoring and control.
- **Onboard LEDs and Push Buttons:** Offer user interaction and visual debugging during testing and development.
- **Peripheral Connector Support:** Headers available for IR sensors, servo motors, ultrasonic sensors, and OLED display.
- **Compact and Student-Friendly Layout:** Optimized for prototyping and academic use, while remaining scalable for more complex designs.
- **Cost-Efficient and Readily Available Components:** Ensures that the board is affordable and easy to reproduce for classroom or lab use.

Chapter 5

APPLICATIONS OF THE ADAPTIVE FPGA BOARD

The adaptive FPGA board has been designed to serve a wide range of applications in embedded systems, robotics, and automation. Because of its flexible I/O, memory support, and robust connectivity, it is suitable for both educational and real-world use cases. FPGAs have been increasingly adopted in domains where deterministic timing, parallel processing, and hardware-level customization are required [5, 9].

Some of the most relevant applications of this board are as follows:

- **Line-Following Robot (LFR):** Using onboard IR sensors, the FPGA can process line contrast and control wheel motors accordingly. FPGAs are particularly well-suited for real-time signal processing required in robotics path planning [9].
- **Obstacle Avoidance System:** With the help of ultrasonic sensors (e.g., HC-SR04), the board can be programmed to detect obstacles and react in real time by altering motor directions or triggering alerts.
- **Robotic Arm Control:** Through PWM generation on the FPGA, users can control servo motors (such as SG90) for precise movement. The hardware-based timing control makes FPGAs better at handling simultaneous axis movement than software-based microcontrollers.
- **Environmental Monitoring System:** The board supports DHT22 sensors for temperature and humidity monitoring, enabling use in smart home or industrial automation settings.
- **IoT Gateway Node:** By integrating with Wi-Fi (ESP8266) or Bluetooth (HC-05) modules through UART or SPI, the FPGA board can function as a local processing and communication node in IoT systems.
- **Data Acquisition and Display System:** Data collected from sensors can be visualized using the connected OLED (SSD1306) display, controlled via I2C from the FPGA, ideal for laboratory test setups.
- **Educational FPGA Trainer Kit:** With all essential interfaces onboard, the board is perfect for teaching students about digital design, interfacing, and embedded system development.

The deterministic and reconfigurable nature of FPGAs makes them highly suitable for embedded control systems, especially in scenarios where flexibility and real-time processing are required [4].

Chapter 6

PIN MAPPING

The pin mapping of the adaptive FPGA board is a critical part of its overall architecture. To establish reliable communication between the FPGA and external peripherals (such as sensors, actuators, and user interfaces), the board features two 80-pin DF40c high-density connectors, designated as **X-CONN** and **Y-CONN**.

In our design, the **male connectors** are soldered onto the adaptive FPGA board, while the corresponding **female connectors** are present on a separate motherboard that carries all the peripheral components—such as IR sensors, servo motors, distance sensors, seven-segment displays, keypads, BLE modules, and more. This physical arrangement allows the adaptive board to be *plugged directly* into the motherboard. The DF40c connectors act as a bridge, allowing every pin on the adaptive board to connect precisely with the component it is intended to control or read from.

Each pin on the male connector is mapped either to an FPGA general-purpose I/O pin or to a dedicated communication, power, or control signal. By referencing the detailed pin mapping tables for both X-CONN and Y-CONN (provided in the following pages), users can quickly identify which peripheral connects to which FPGA pin. This structured mapping supports modular development and ensures long-term maintainability and scalability.

Beyond the X-CONN and Y-CONN mappings, several other connections are also established on the adaptive board:

- **Power Management:** Multiple voltage domains are supported across the board. Pins are allocated for **VCCINT (1.3V)**, **VCCAUX (1.8V)**, and **VCCO (3.3V)**. These voltages power both the FPGA core and the I/O banks. Ground connections (GND) are distributed throughout both connectors to ensure stable power delivery and signal integrity.
- **Communication Interfaces:**
 - **SPI Interface:** Includes SCK, MISO, and MOSI lines routed to the FPGA for communicating with devices like LoRa modules, flash memory, or sensor hubs.
 - **UART Lines:** RX and TX lines are connected for BLE (HC-05) and Wi-Fi (ESP8266) modules, enabling wireless communication.
 - **I²C Bus:** SDA and SCL pins are provided for interfacing with devices such as OLED displays or real-time clocks.

- **Input/Output Peripherals:**

- Multiple GPIO pins are assigned to LEDs (LED1–LED16), push buttons (PB1–PB5), and switches (SW1–SW16).
- PWM-capable GPIOs are used for controlling servo motors and for generating signals for the motor driver's IN and EN pins.

- **Reset and Clocking:**

- A dedicated reset pin (CPU_RST) is connected to a push button to reset the FPGA.
- An external clock pin (CLK) is allocated for synchronous logic and communication protocols like SPI.

- **Analog and Mixed-Signal Interfaces:**

- ADC reference voltage pins (ADC VREF+ and VREF-) are mapped for precise analog measurements from sensors such as LDR, vibration sensors, and microphones.

This comprehensive pin mapping strategy ensures that the board can be used across a wide range of embedded systems and robotics projects without needing redesigns for each application. By simply plugging the adaptive board into a motherboard carrying the necessary peripherals, the system becomes ready for programming, prototyping, or deployment.

Below are the detailed pin mappings for both X-CONN and Y-CONN connectors:

6.1 X-CONN Pin Mapping

Female Connector Pin	Male Connector Pin	Peripheral name	FPGA PIN
X1	X1	Keypad R1	GPIO_1
X2	X2	Keypad R2	GPIO_2
X3	X3	Keypad R3	GPIO_3
X4	X4	Keypad R4	GPIO_4
X5	X5	Keypad C1	GPIO_5
X6	X6	Keypad C2	GPIO_6
X7	X7	Keypad C3	GPIO_7
X8	X8	Keypad C4	GPIO_8
X9	X9	SPI SCK	CLK
X10	X10	SPI MISO	MISO
X11	X11	SPI MOSI	MOSI
X12	X12	HC05 BLE RXD	FPGA_RX
X13	X13	HC05 BLE TXD	FPGA_TX
X14	X14	ESP8266 WLAN RXD	FPGA_RX
X15	X15	ESP8266 WLAN TXD	FPGA_TX
X16	X16	GPIO 1	X
X17	X17	GPIO 2	X
X18	X18	GPIO 3	X
X19	X19	GPIO 4	X
X20	X20	NC	NC
X21	X21	VIN	VCCINT
X22	X22	GND	GND_FPGA_5
X23	X23	NC	NC
X24	X24	LED 1	GPIO_24
X25	X25	LED 2	GPIO_25
X26	X26	LED 3	GPIO_26
X27	X27	GPIO 5	X
X28	X28	GPIO 6	X
X29	X29	GPIO 7	X
X30	X30	GPIO 8	X
X31	X31	GND	GND_FPGA_4
X32	X32	VIN	VCCAUX
X33	X33	LED 4	GPIO_33
X34	X34	LED 5	GPIO_34
X35	X35	LED 6	GPIO_35
X36	X36	LED 7	GPIO_36
X37	X37	LED 8	GPIO_37
X38	X38	LED 9	GPIO_38
X39	X39	LED 10	GPIO_39
X40	X40	LED 11	GPIO_40
X41	X41	VIN	VCCINT
X42	X42	GND	GND_FPGA_3

Female Connector Pin	Male Connector Pin	Peripheral name	FPGA PIN
X43	X43	NC	NC
X44	X44	I2C SDA	GPIO_44
X45	X45	NC	NC
X46	X46	I2C SCL	GPIO_46
X47	X47	GND	GND_FPGA_2
X48	X48	VIN	VCCAUX
X49	X49	LED 12	GPIO_49
X50	X50	LED 13	GPIO_50
X51	X51	LED 14	GPIO_51
X52	X52	LED 15	GPIO_52
X53	X53	VSYS	VCCO
X54	X54	GND	GND_FPGA_1
X55	X55	NC	NC
X56	X56	Switch 1	GPIO_56
X57	X57	Switch 2	GPIO_57
X58	X58	Switch 3	GPIO_58
X59	X59	Switch 4	GPIO_59
X60	X60	Switch 5	GPIO_60
X61	X61	Switch 6	GPIO_61
X62	X62	Switch 7	GPIO_62
X63	X63	Switch 8	GPIO_63
X64	X64	VSYS	VCCO
X65	X65	Switch 9	GPIO_65
X66	X66	Switch 10	GPIO_66
X67	X67	Switch 11	GPIO_67
X68	X68	Switch 12	GPIO_68
X69	X69	Switch 13	GPIO_69
X70	X70	Switch 14	GPIO_70
X71	X71	Switch 15	GPIO_71
X72	X72	Switch 16	GPIO_72
X73	X73	RESET	CPU_RST
X74	X74	NC	NC
X75	X75	Push button 1	GPIO_75
X76	X76	Push button 2	GPIO_76
X77	X77	Push button 3	GPIO_77
X78	X78	Push button 4	GPIO_78
X79	X79	NC	NC
X80	X80	NC	NC

6.2 Y-CONN Pin Mapping

Female Connector Pin	Male Connector Pin	Peripheral name	FPGA PIN
Y1	Y1	Color sensor S0	GPIO_Y1
Y2	Y2	Moisture sensor signal	GPIO_Y2
Y3	Y3	Color sensor S1	GPIO_Y3
Y4	Y4	Color sensor S2	GPIO_Y4
Y5	Y5	GPIO 9	X
Y6	Y6	GPIO 10	X
Y7	Y7	GPIO 11	X
Y8	Y8	GPIO 12	X
Y9	Y9	GPIO 13	X
Y10	Y10	GPIO 14	X
Y11	Y11	GPIO 15	X
Y12	Y12	GPIO 16	X
Y13	Y13	Buzzer	GPIO_Y13
Y14	Y14	Color sensor S3	GPIO_Y14
Y15	Y15	Seven segment display (a)	GPIO_Y15
Y16	Y16	Seven segment display (b)	GPIO_Y16
Y17	Y17	Seven segment display (c)	GPIO_Y17
Y18	Y18	Seven segment display (d)	GPIO_Y18
Y19	Y19	Seven segment display (e)	GPIO_Y19
Y20	Y20	Seven segment display (f)	GPIO_Y20
Y21	Y21	Seven segment display (g)	GPIO_Y21
Y22	Y22	Color sensor Out	GPIO_Y22
Y23	Y23	VCC / +3V3	VCCO
Y24	Y24	GND	GND_FPGA_1
Y25	Y25	Seven segment display (a1)	GPIO_Y25
Y26	Y26	Seven segment display (a2)	GPIO_Y26
Y27	Y27	Seven segment display (a3)	GPIO_Y27
Y28	Y28	Seven segment display (a4)	GPIO_Y28
Y29	Y29	Seven segment display (a5)	GPIO_Y29
Y30	Y30	Seven segment display (a6)	GPIO_Y30
Y31	Y31	Seven segment display (a7)	GPIO_Y31
Y32	Y32	Seven segment display (a8)	GPIO_Y32
Y33	Y33	GND	GND_FPGA_2
Y34	Y34	VCC	VCCINT
Y35	Y35	NC	NC
Y36	Y36	Motor driver IN2	GPIO_Y36
Y37	Y37	NC	NC
Y38	Y38	Left IR sensor pin	GPIO_Y38
Y39	Y39	NC	NC
Y40	Y40	Right IR sensor pin	GPIO_Y40
Y41	Y41	NC	NC
Y42	Y42	Motor driver IN4	GPIO_Y42

Female Connector Pin	Male Connector Pin	Peripheral name	FPGA PIN
Y43	Y43	VCC / +3V3	VCCO
Y44	Y44	GND	GND_FPGA_3
Y45	Y45	NC	NC
Y46	Y46	PIR motion sensor	GPIO_Y46
Y47	Y47	NC	NC
Y48	Y48	LoRa G0	GPIO_Y48
Y49	Y49	LoRa CS	GPIO_Y49
Y50	Y50	LoRa RST	GPIO_Y50
Y51	Y51	RFID CS	X
Y52	Y52	RFID RST	X
Y53	Y53	NC	NC
Y54	Y54	DHT22 temperature sensor pin	X
Y55	Y55	NC	NC
Y56	Y56	NC	NC
Y57	Y57	GND	GND_FPGA_4
Y58	Y58	NC	NC
Y59	Y59	Distance sensor trigger	D6
Y60	Y60	Motor driver IN1	D1
Y61	Y61	Left servo signal	D5
Y62	Y62	Motor driver EnA	D0
Y63	Y63	Motor driver IN3	D4
Y64	Y64	NC	NC
Y65	Y65	Motor driver EnB	D3
Y66	Y66	NC	NC
Y67	Y67	Right servo signal	D2
Y68	Y68	NC	NC
Y69	Y69	VCC / +3V3	VCCO
Y70	Y70	GND	GND_FPGA_5
Y71	Y71	ADC VREF +	X
Y72	Y72	ADC VREF -	X
Y73	Y73	LDR	A0
Y74	Y74	Piezo vibration sensor	A4
Y75	Y75	Potentiometer 2	A1
Y76	Y76	Distance sensor echo	D20
Y77	Y77	LED 16	A2
Y78	Y78	Push button 5	D21
Y79	Y79	Microphone analog pin	A3
Y80	Y80	NC	NC

Chapter 7

PERIPHERALS

This section details the critical peripherals designed into the adaptive FPGA board. Each component is selected to maximize modularity, performance, and suitability for reconfigurable and embedded system applications.

Power Management Circuit

The power management subsystem is responsible for regulating, distributing, and protecting voltage supplies to the various blocks on the FPGA board. It typically includes low-dropout voltage regulators (LDOs), Schottky diodes, and decoupling capacitors. This design enables the board to derive multiple regulated rails—such as 3.3V for I/O, 1.2V for FPGA core, and 2.5V for auxiliary peripherals—from a single USB 5V input or external supply. Power sequencing is critical: the FPGA core voltage must be stabilized before auxiliary and I/O voltages are enabled, to prevent latch-up or undefined behavior during configuration. In your adaptive board, this stable power foundation ensures safe operation of high-speed interfaces like DDR2 and SPI, and also protects the FPGA fabric during boot and resets. A power-good LED is connected to the 3.3V line to indicate readiness, assisting in visual verification during testing or production. Without a stable power subsystem, the FPGA logic may misconfigure or fail to boot entirely [2].

High-Density DF40C Connectors

Your board includes two DF40C 80-pin high-density board-to-board connectors that form the central bridge between the adaptive FPGA module and the external motherboard. These connectors expose nearly all general-purpose I/Os, communication lines, and power rails from the FPGA to the external system. This modular approach allows the same FPGA core to interface with different hardware subsystems—such as robotics controllers, sensors, or actuators—by simply plugging into a different baseboard. The DF40C series from Hirose is known for high-speed signal support, compact pitch (0.4 mm), and robust mechanical locking. The connectors make the board ideal for stacking applications and ensure impedance-controlled routing suitable for DDR2, SPI, and other synchronous interfaces. In this project, the connectors provide the main pathway for SPI-controlled OLED displays, I2C sensor arrays, IR-based obstacle sensors, or even actuator control via UART-based motor drivers [10].

JTAG Interface

The Joint Test Action Group (JTAG) interface is essential for in-system configuration and debugging of the FPGA. It enables boundary-scan testing, logic configuration through bitstream upload, and post-configuration signal analysis. In your design, a standard 6-pin JTAG header is connected directly to the Spartan-7 FPGA and routed to the top layer for easy access. This interface works with tools like Xilinx Vivado and iMPACT to program the board and observe internal logic. For instance, when verifying the DDR2 controller or the SPI communication core, the JTAG allows you to set breakpoints, toggle test inputs, or read internal registers using Chipscope or ILA (Integrated Logic Analyzer) IPs. The presence of JTAG simplifies the development cycle and eliminates the need for external flash or USB interfaces during firmware prototyping [16].

USB-UART and SPI Communication Interfaces

The board includes dual communication interfaces: USB-UART and SPI. The USB-UART bridge (such as a CP2102 or FT2232) connects the FPGA to a host PC via a virtual COM port, allowing real-time data exchange, logging, and user input through terminal applications like PuTTY. In your case, this is especially useful for logging events like configuration status, memory usage, or sensor data from the external motherboard. SPI (Serial Peripheral Interface), on the other hand, is a synchronous, full-duplex protocol ideal for interfacing with OLED displays, shift registers, and ADCs. On your adaptive board, the SPI lines are routed through the DF40C connector, enabling modular attachment to SPI-based peripherals like OLED or LoRa modules. The separation of control (UART) and sensor-actuator communication (SPI) ensures modularity, clarity in signal flow, and robust debugging [12, 7].

Memory Module Support

The board supports external DDR2 memory via a dedicated memory interface connected to the Spartan-7. The DDR2 SDRAM allows high-speed, large-volume data storage and buffering, which is essential for applications requiring frame buffering, temporary sensor data storage, or real-time signal processing. For example, in a use case where the FPGA is controlling a display or capturing sensor readings at high frequency, internal BRAM is insufficient. The DDR2 interface is implemented using the Xilinx MIG (Memory Interface Generator) IP, which handles timing constraints, clock domain synchronization, and latency management. This memory system enables your project to scale into more complex workloads, including lightweight video processing, packet buffering in IoT gateways, or real-time motor feedback systems [13].

LED Indicator and Reset Buttons

The board includes a single LED and two push buttons. The LED is connected to the 3.3V line through a resistor and serves as a power indicator. This is useful for basic health checking and user feedback, especially when operating headless (without PC connection). One push button is wired to the configuration reset pin, enabling reloading of the FPGA bitstream during debugging or recovery. The second button performs an FPGA CPU reset, which can be used to restart logic cores like MicroBlaze or user-defined FSMs without reprogramming the entire chip. Both buttons are active-low and debounced either through hardware (RC network) or logic (FPGA). In your modular system, these buttons help reset the board before re-attaching to new daughterboards or clearing memory states during demos [11, 8].

Crystal Oscillator

A 50 MHz crystal oscillator is included on the board to provide a clean, stable reference clock for the FPGA. The Spartan-7's global clock input receives this signal and may use MMCMs (Mixed-Mode Clock Managers) or PLLs to derive secondary clocks (e.g., 100 MHz for memory, 25 MHz for SPI). Clock stability is critical for synchronous systems where timing violations can lead to metastability or data corruption. Your oscillator ensures reliable timing for DDR2 memory access, serial protocol timings, and logic state transitions. For example, when reading from sensors over SPI and writing to DDR2 buffers, synchronized and phase-aligned clocking ensures data coherency. The crystal is selected for low jitter and startup delay, suitable for both industrial and academic use [1].

FPGA Core: Xilinx Spartan-7

The computational engine of the board is the Xilinx Spartan-7 XC7S25-1FTGB196C FPGA. This 28nm CMOS programmable logic device offers 23,360 logic cells, 120 user I/Os, multiple block RAMs (BRAM), and dedicated DSP slices. It also includes integrated clock managers, high-speed serial transceivers, and support for MicroBlaze soft-core processors. In your adaptive system, this FPGA enables custom digital logic for tasks such as real-time sensor data processing, protocol bridging, and motor control. Its reconfigurability allows reuse across multiple applications—robotic control, IoT edge computation, or educational labs—by reprogramming the logic fabric. The Spartan-7 strikes a balance between cost, power, and capability, making it ideal for modular, reusable FPGA projects like yours [15].

Chapter 8

SCHEMATIC AND PCB OVERVIEW

This chapter provides a comprehensive description of the schematic design and PCB layout implemented for the adaptive FPGA board. The schematic is divided into functional blocks, each of which serves a specific purpose in the electrical and system-level design. These blocks are discussed below in the order they appear in the complete system.

The first subsystem is the **Power Management and Voltage Regulation Block**, which supplies the core voltage rails required by the FPGA and auxiliary components. The design includes low-dropout voltage regulators that convert a 5V USB input into stable 3.3V and 1.2V outputs. Proper decoupling capacitors are placed near each power pin, and the 3.3V line is connected to an indicator LED that signals the board's power status. Input protection is provided by reverse-polarity diodes and inrush current-limiting resistors. This block ensures voltage stability and startup safety and lays the electrical foundation for all downstream modules. This block is shown in the Appendix as Figure 8.1.

The **USB-UART and SPI Communication Block** combines two vital communication protocols. The USB-UART section allows serial communication between the host computer and the FPGA, enabling terminal-based debugging and firmware logging. The SPI section enables the FPGA to control peripherals such as OLED displays and sensors. Pull-up resistors are included for SPI signal integrity, and all lines are routed through the DF40C connectors to external modules. The combined communication subsystem provides a flexible interface for both control and data flow. You can see the corresponding schematic in Figure 8.2 in the Appendix.

The **JTAG Interface Block** provides the programming and debugging entry point to the FPGA. It includes a 6-pin standard JTAG connector wired directly to the Spartan-7's configuration pins. This block allows the user to upload bitstreams, perform boundary-scan testing, and use Xilinx debugging tools like Vivado's Integrated Logic Analyzer. Proper signal terminations ensure clean JTAG clocking and reliable operation during configuration. This block is illustrated in Figure 8.3 in the Appendix.

The **Memory Module Block** includes routing for external DDR2 SDRAM, although the module is optional in this version. Control lines, address buses, and data I/Os are all grouped for synchronous timing and routed from the FPGA to the memory footprint. The goal of this module is to support data buffering for high-throughput applications in future versions of the board. The schematic layout for this module is shown in Figure 8.4 in the Appendix.

The **Reset and LED Block** features two push-buttons and one indicator LED. One push-button is connected to the FPGA configuration reset pin, and the other to the system reset input for internal soft-core processors like MicroBlaze. The LED is connected through a resistor to the 3.3V rail and confirms power delivery visually. These I/O elements are essential for manual testing, debugging, and system reinitialization. This schematic can be reviewed in Figure 8.5 in the Appendix.

The **Clock Generation Block** consists of a 50 MHz crystal oscillator connected to the FPGA's global clock input. This stable reference clock feeds the Mixed-Mode Clock Manager (MMCM) inside the Spartan-7, which can then generate other required clock frequencies for different modules. Proper load capacitors and series resistors are included to ensure stable startup and reduce phase noise. This subsystem is critical for synchronous design and is depicted in Figure 8.6 in the Appendix.

The **DF40C Connector Block** implements the two 80-pin Hirose DF40C male connectors that allow the adaptive FPGA board to plug into a larger motherboard. All communication signals (SPI, UART, I2C), GPIOs, and power rails are routed through these connectors. The horizontal spacing between the connectors is 43.18 mm, matching the layout of the Arduino Portenta breakout board, ensuring mechanical and electrical compatibility. These connectors are the interface backbone that enables the board's adaptive capabilities. The schematic is included in Figure 8.7 in the Appendix.

Lastly, the **FPGA Core Block** features the Xilinx Spartan-7 XC7S25 FPGA in an FTGB196 BGA package. It is connected to three separate power domains: VCCINT (1.2V core voltage), VCCAUX (2.5V auxiliary), and VCCIO (3.3V I/O). The block also includes all necessary configuration, global clock, and system reset lines. Most of the general-purpose I/O pins are routed through the DF40C connectors, allowing the board to adapt to various external peripheral arrangements. Proper pull-up and pull-down resistors are added to critical configuration pins such as INIT_B, DONE, and PROGRAM_B to ensure stable boot behavior and avoid floating logic states. This block serves as the heart of the adaptive system, enabling full reprogrammability and user-defined logic. The complete schematic for the FPGA core can be found in Figure 8.8 in the Appendix.

Following the completion of the schematic design, the next phase in the development process was the layouting. The transition from schematic to layout required careful consideration of electrical connectivity, mechanical constraints, component placement, and signal integrity. The board layout was created to house the Xilinx Spartan-7 XC7S25 FPGA along with all supporting peripherals, including power regulation circuits, communication interfaces, memory, clock sources, user I/O, and high-density connectors.

The first step in PCB layout involved defining the board outline and establishing the mechanical constraints dictated by the two DF40C high-density connectors. These connectors were placed

precisely 43.18 mm apart to ensure compatibility with an external motherboard, similar to the Arduino Portenta breakout board. Mounting holes and connector footprints were then placed in accordance with the schematic, followed by strategic arrangement of core components such as voltage regulators, the JTAG interface, memory pads, and oscillator modules.

Once the component placement was finalized, the board stack-up was defined. Unlike simpler designs that rely on 2- or 4-layer PCBs, this adaptive FPGA board was implemented on a high-density **10-layer PCB**. This multi-layer configuration allowed for optimized routing, dedicated power and ground planes, and controlled impedance paths for high-speed signals. The stack-up included alternating signal layers with interleaved power and ground planes to reduce electromagnetic interference (EMI) and crosstalk while maintaining electrical isolation between critical domains such as VCCINT, VCCAUX, and VCCIO.

The routing of the board was done using KiCad's built-in routing tool, following custom rules for trace width, spacing, and via sizes. Important high-speed connections—like the ones used for the DDR2 memory and clock signals—were routed with carefully measured lengths to ensure they remain synchronized. In some cases, zig-zag or meandered traces were used to match signal timing. For signals that require high precision, like USB data lines, the traces were routed as balanced pairs on internal layers to reduce noise and maintain reliable communication.

The use of a ball-grid array (BGA) package for the Spartan-7 FPGA introduced significant routing challenges. With 196 balls and a high number of general-purpose I/O (GPIO) signals, careful fan-out planning was essential. Blind, buried, and microvias were employed to escape inner balls and to reduce congestion under the BGA. Several inner layers were allocated specifically to facilitate this routing, providing additional channels for signals that could not be routed on outer layers. The inclusion of such via types enabled successful breakout and routing of all necessary FPGA signals to their corresponding blocks.

In addition to signal routing, several layers were reserved as solid planes for power distribution and grounding. This provided a low-impedance path for power delivery and improved thermal performance. Decoupling capacitors were placed as close as possible to the power pins of the FPGA and other active components, following standard high-speed design practices. Special attention was given to the oscillator block and clock lines, which were routed using short, shielded paths to minimize jitter and phase noise.

The visual inspection and review were complemented by 3D renderings of the board, which provided spatial verification of component heights, mounting hole clearance, and connector alignment. Screenshots of these 3D models are presented in Figures 8.9 to 8.13 in the Appendix for visual confirmation.

A major challenge faced during the PCB phase was the **routing congestion caused by the dense peripheral configuration and the complexity of the FPGA chip**. The abundance of signals, tight component spacing, and need for signal integrity required extensive layer planning and iterative routing. This phase was the most time-consuming and technically demanding part of the design, as even minor misroutes could cause timing issues, configuration failures, or mechanical misalignment.

Despite the challenges, the board was successfully routed, reviewed, and exported as Gerber files for manufacturing. The result is a professionally laid out, multi-layer PCB that meets the mechanical and electrical demands of an adaptive, reprogrammable embedded system. The design reflects a careful balance between technical precision and practical limitations of prototype development within the academic timeline.

Chapter 9

EVALUATION

After completing the PCB layout in KiCad, several evaluation steps were undertaken to validate both the electrical correctness and mechanical compatibility of the adaptive FPGA board. The first step involved running the built-in Design Rule Check (DRC) tool in the KiCad PCB editor. The DRC report, as shown in Figure 9.1, confirmed zero errors in the design, indicating that the trace widths, clearances, and via dimensions all adhered to the electrical constraints. Although some warnings were generated, mostly concerning the silkscreen layer, such as overlapping labels or proximity of silkscreen text to the board edge, these were acknowledged but not resolved due to time constraints, as they do not impact functionality or manufacturability.

Following the DRC validation, one of the most critical evaluations involved verifying the placement and alignment of the two high-density 80-pin DF40C male connectors on the bottom side of the adaptive FPGA board. These connectors form the primary physical and electrical bridge between the FPGA module and a separate motherboard that hosts peripherals like sensors, actuators, and communication bridges. To ensure successful interfacing, the connectors must match exactly with corresponding female connectors on the motherboard, both in horizontal spacing and vertical alignment. Any deviation in position could lead to mechanical failure or unreliable electrical contact.

To guide this mechanical alignment, the layout of the Arduino Portenta H7 breakout board was used as a dimensional reference. Although this project does not involve the Arduino board itself, the Portenta breakout board was chosen due to its documented and publicly available dimensions. According to the official documentation [3], the high-density connectors J1 and J2 on the breakout board are positioned at 17.14 mm and 60.32 mm, respectively, from the left edge of the PCB. Subtracting these values gives a connector spacing of 43.18 mm, which was then adopted in this design. This specific spacing ensures that the adaptive FPGA board can align with existing motherboard layouts that follow the same connector standard.

This 43.18 mm spacing was manually verified in the PCB layout by placing a dimension line between the connectors using the User.Drawings layer in KiCad. Additionally, the vertical alignment was also verified by ensuring both connectors shared the same Y-axis coordinate. Proper vertical alignment is crucial to avoid pin mismatch or physical stress during insertion into the motherboard.

This evaluation confirms the adaptive capability of the board. Once the FPGA is programmed

with the correct bitstream, it can be inserted into the motherboard and immediately recognize and communicate with any connected peripherals via the DF40C connectors. The FPGA is configured to detect the data paths and act accordingly, making the board highly modular and reusable across different applications. The successful alignment of these connectors, along with a clean DRC result, verifies the integrity of the PCB design and validates its use in real-world adaptive embedded systems.

Finally, all electrical connections implemented in both the schematic and PCB layout were carefully designed in accordance with the official datasheets and reference manuals of the electronic components used. This includes the pin mappings, voltage supply lines, decoupling requirements, signal terminations, and communication interfaces. Adherence to component specifications ensures reliable electrical behavior and system stability under real operating conditions. This approach also minimizes the risk of hardware failures during integration with the motherboard and supports the long-term reusability of the adaptive FPGA module across multiple applications.

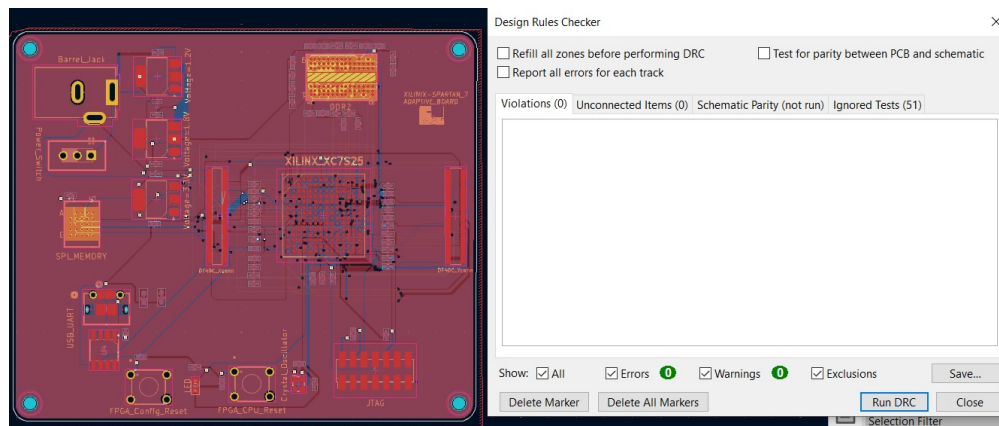


Figure 9.1: Design Rule Check completed successfully.

Chapter 10

FUTURE WORK

Although the adaptive FPGA board developed in this project successfully fulfills the functional and mechanical requirements set at the beginning, there remains room for further enhancement and refinement in future iterations. The current design, implemented using a standard 10-layer PCB stack-up, was sufficient to route all required signals and power lines within the constraints of the project. However, a transition to additional layers if high complex system is used so that would significantly improve signal integrity, reduce EMI (electromagnetic interference), and allow for cleaner separation of analog, digital, and power domains. With additional layers, dedicated planes for ground and power (e.g., 3.3V or 1.2V) could be introduced, enabling impedance-controlled routing for high-speed interfaces like SPI and DDR2.

Another area of improvement involves optimizing the placement and routing of high-speed signals. While the current routing strategy respects design rules and functions correctly, differential pair signals or clock lines could benefit from better shielding and tighter impedance matching. This becomes especially relevant in applications where the adaptive board is used in noisy environments or paired with complex peripheral boards.

Mechanically, the DF40C connector alignment is accurate and functional, but future versions of the adaptive board could implement keyed or chamfered edges to simplify insertion and prevent reverse mounting. Additionally, mechanical standoffs or mounting holes could be added to improve structural stability when the board is plugged into a motherboard, particularly in industrial or robotic applications where vibrations are common.

Functionally, one promising future direction is the integration of an onboard non-volatile memory such as a QSPI flash chip. This would allow the FPGA to store and load bitstreams independently without relying on an external programmer or JTAG interface, thereby enabling field reconfiguration and plug-and-play operation. Combined with an embedded MicroBlaze processor, the FPGA could be configured to auto-detect connected peripherals through initialization protocols (e.g., I2C device scanning or GPIO probing) and dynamically adjust its logic configuration at runtime.

Another possible extension is the inclusion of test points or debug headers for easier signal probing and logic analysis during development. While the current board is streamlined and compact, accessibility to key signals would support faster validation, especially during integration with varied motherboards.

Finally, the adaptiveness of the board can be extended further by defining a common interface specification or "adaptive pin map" for all compatible motherboards. This would allow the same FPGA board to be used across a wide variety of carrier boards without requiring logic changes, reinforcing the reusability and scalability of the design. In summary, while the current implementation is functionally and mechanically validated, future iterations can benefit from improvements in signal integrity, mechanical robustness, self-reconfiguration capabilities, and enhanced developer accessibility. These enhancements would not only improve performance and usability but also elevate the board's value in real-world prototyping and deployment environments.

Chapter 11

TOOLS AND SOFTWARE USED

The development of the adaptive FPGA board involved the use of several tools and software environments, each contributing to different stages of the design, evaluation, and research process. These tools enabled efficient circuit design, layout optimization, verification, and structured documentation.

The primary tool used for schematic capture and PCB layout was KiCad. KiCad was selected for its powerful yet accessible feature set, support for custom libraries, and compatibility with industry-standard manufacturing formats such as Gerber and drill files. The schematic editor was used to implement the full circuit diagram of the adaptive FPGA board, ensuring proper symbol association, electrical rule checking, and hierarchical block organization. The PCB layout editor enabled the routing of all signal and power traces, placement of the DF40C high-density connectors, and the implementation of DRC (Design Rule Check) rules. The User Drawings layer in KiCad also proved useful in dimensionally verifying connector spacing based on the Arduino Portenta breakout board reference.

In addition, ChatGPT was used to support research throughout the project—particularly in selecting a suitable FPGA model, evaluating communication interface options, and reviewing best practices in embedded system design. It also provided support for technical formatting in LaTeX, assisting in structuring and maintaining consistent style across the report chapters. This made it easier to focus on design and evaluation tasks without spending excess time resolving format-related challenges.

Finally, To manage design files and maintain a clear revision history, Git was used for version control. This ensured that schematic changes, PCB layout adjustments, and LaTeX source files were consistently tracked. Such an approach enhanced project reproducibility, simplified collaboration, and provided a secure backup of all design milestones.

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AFFIDAVIT

I, Kashif Raza, herewith declare that I have composed the present paper and work by myself and without use of any other than the cited sources and aids. Sentences or parts of sentences quoted literally are marked as such; other references with regard to the statement and scope are indicated by full details of the publications concerned. The paper and work in the same or similar form has not been submitted to any examination body and has not been published. This paper was not yet, even in part, used in another examination or as a course performance.

Lippstadt, 31 July 2025

Signature *Kashif*

APPENDIX

This appendix includes all schematic and PCB design images referenced throughout the document. These figures provide a visual representation of the implementation discussed in the corresponding chapters.

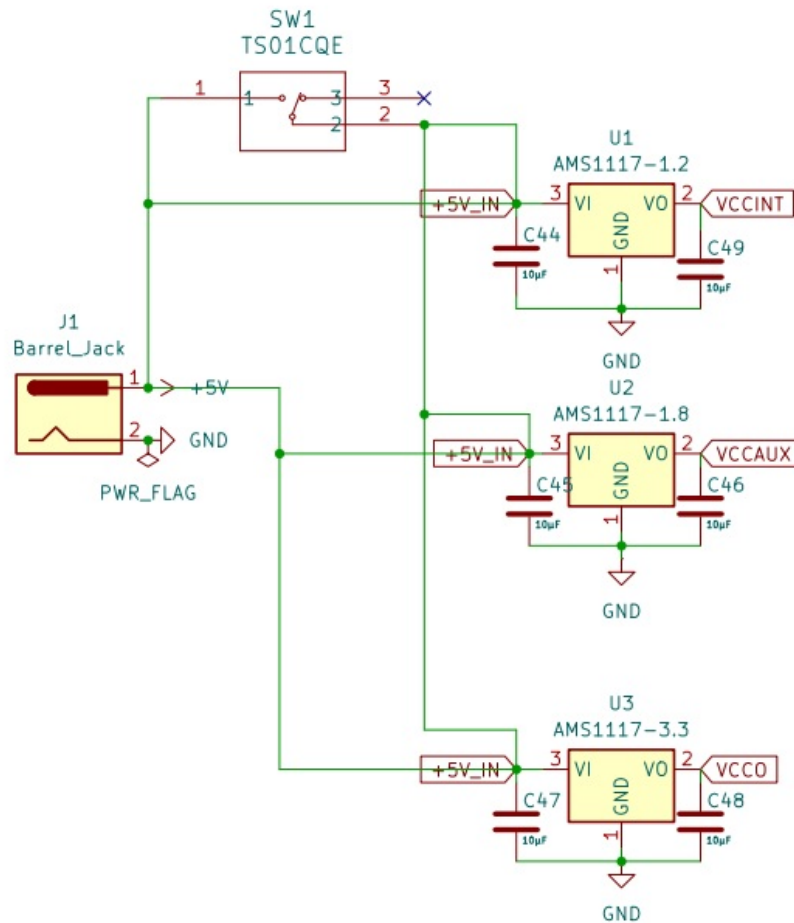


Figure 8.1: Schematic: Power Management and Voltage Regulation Block

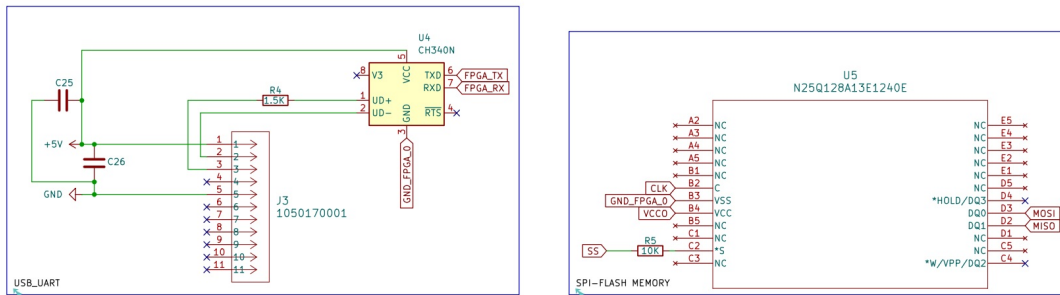


Figure 8.2: Schematic: USB-UART and SPI Communication Interface Block

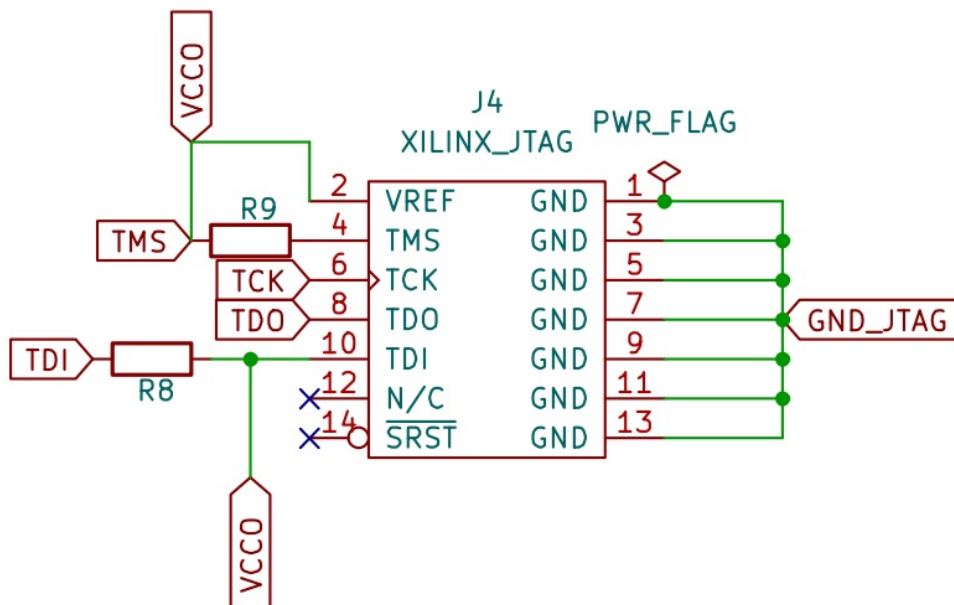


Figure 8.3: Schematic: Xilinx JTAG Configuration Interface Block

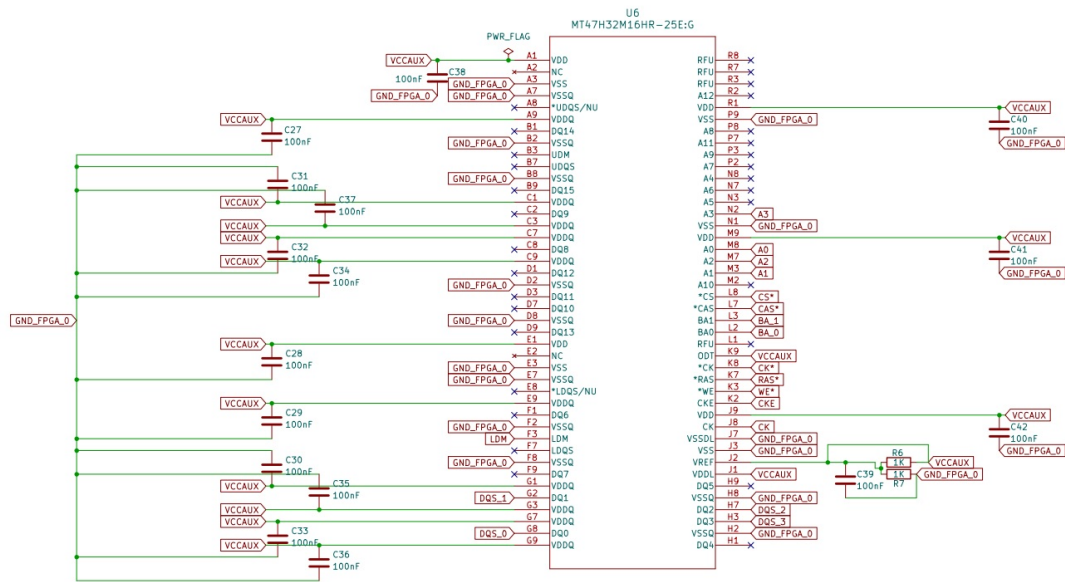


Figure 8.4: Schematic: External Memory Module Support Block

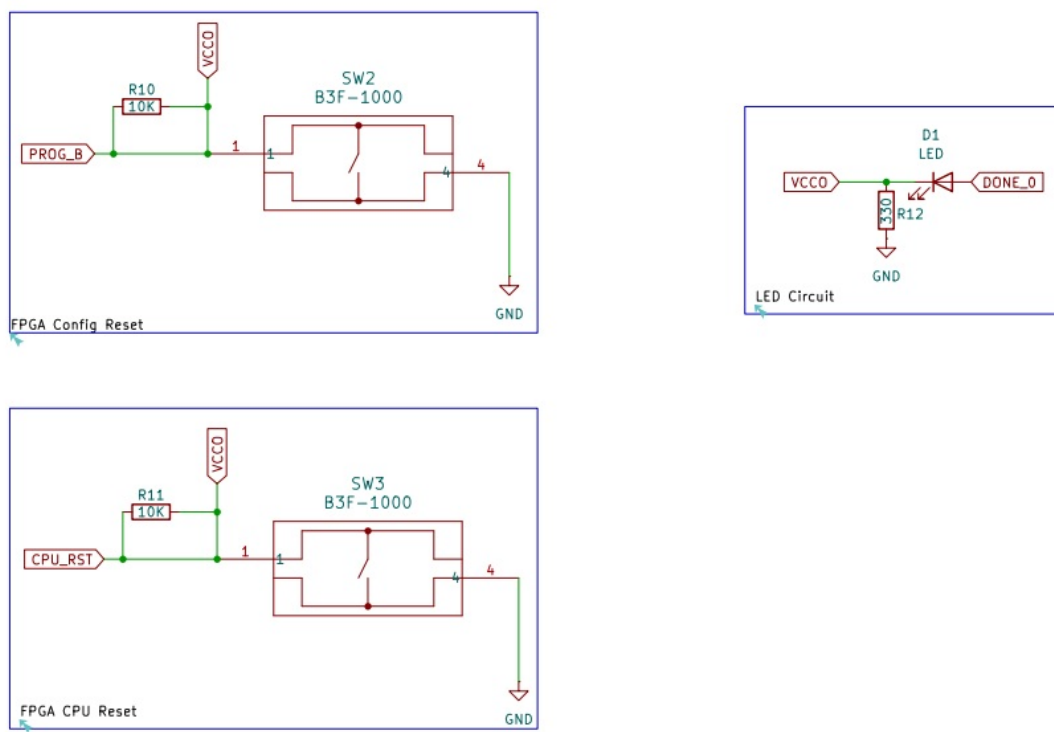


Figure 8.5: Schematic: Reset and Power LED Indicator Block

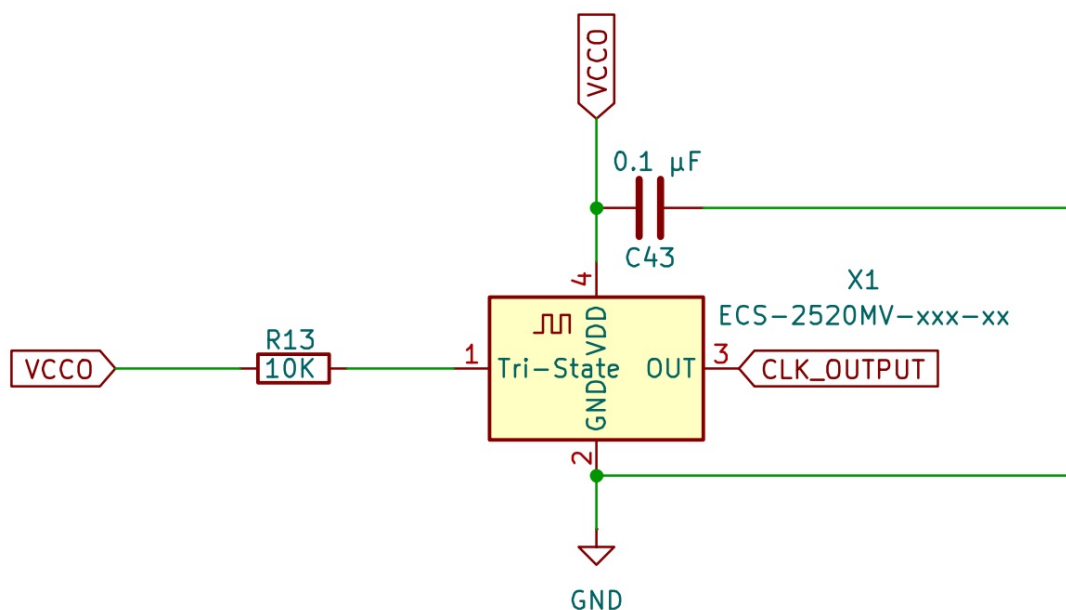


Figure 8.6: Schematic: Clock Generation with Crystal Oscillator

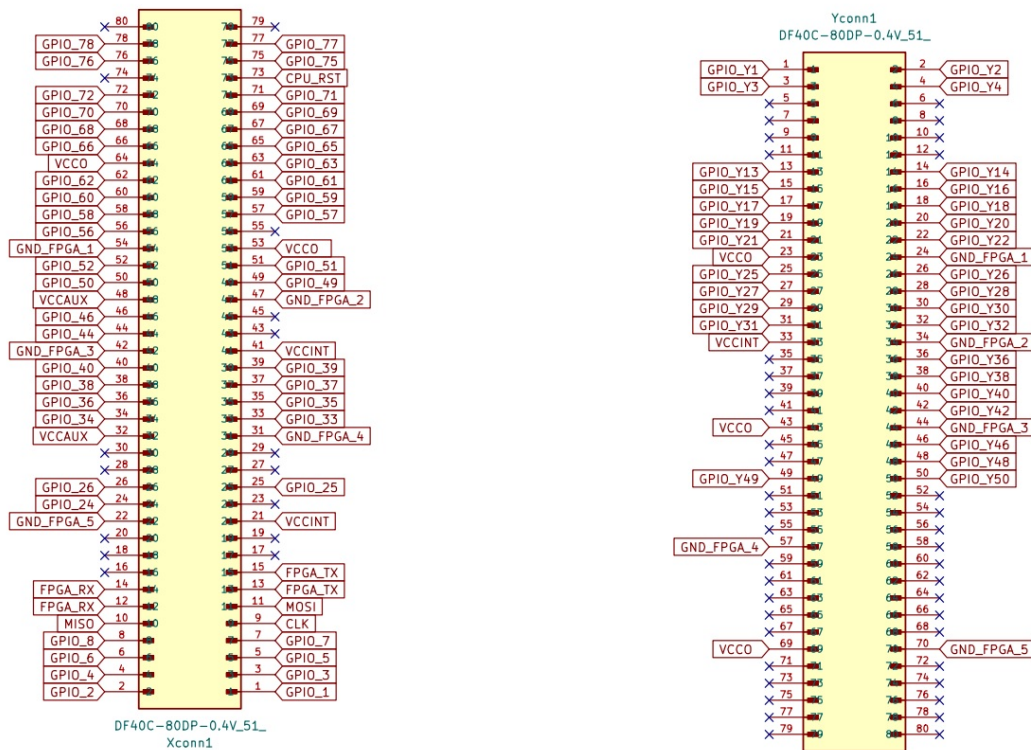


Figure 8.7: Schematic: DF40C High-Density Connector Interface

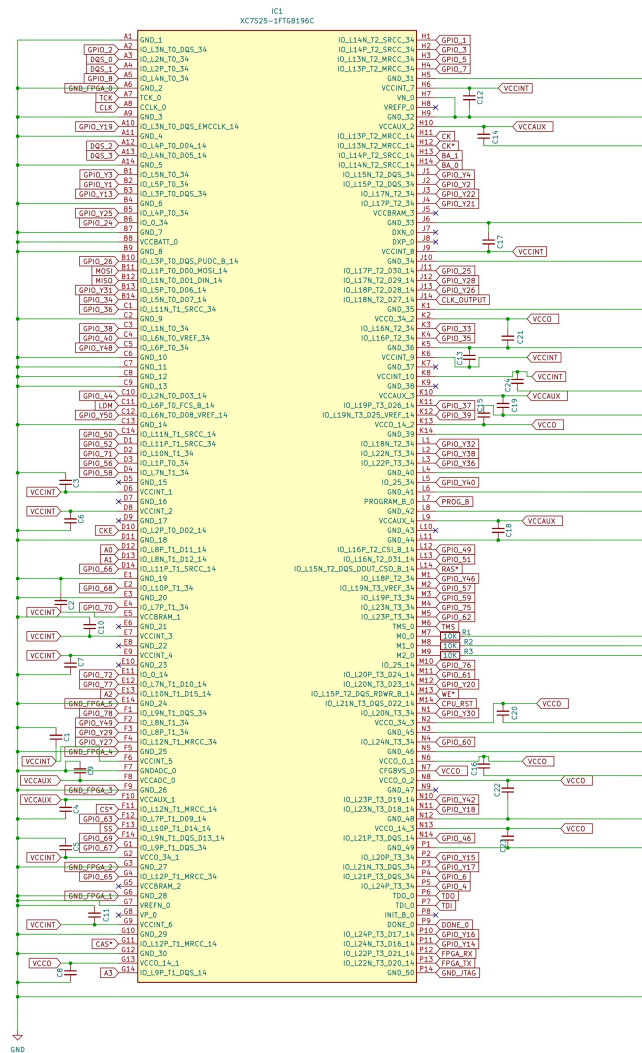


Figure 8.8: Schematic: Xilinx Spartan-7 FPGA Core Block

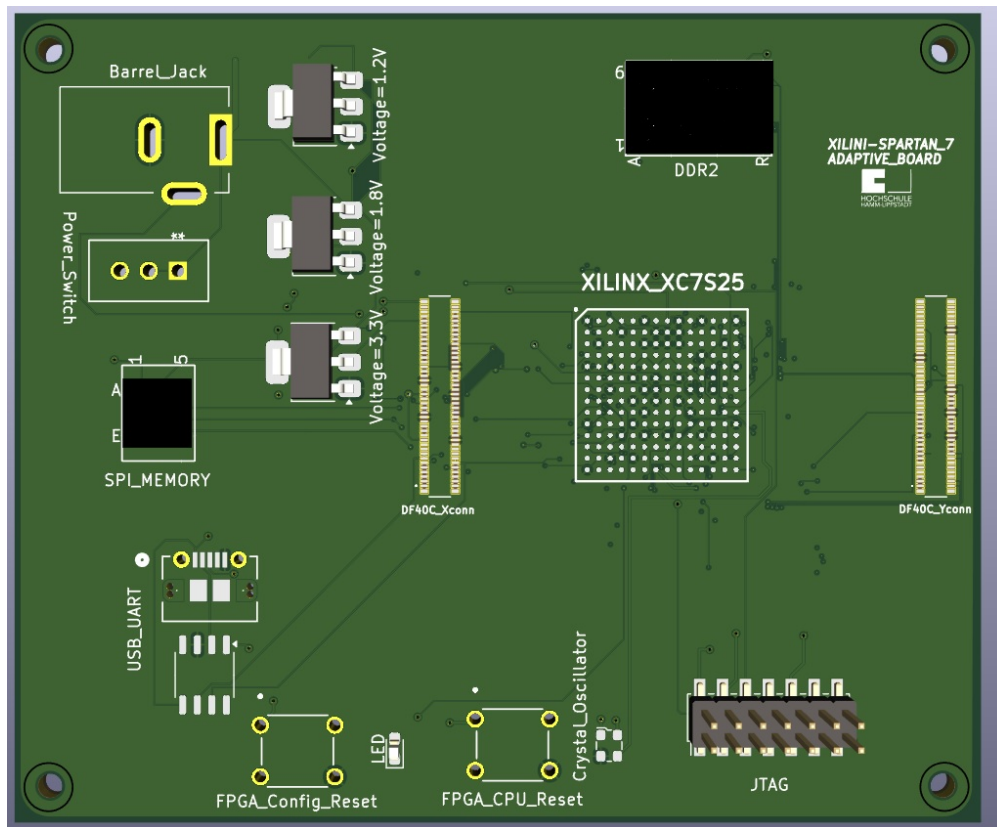


Figure 8.9: 3D View of Final PCB - Front Side

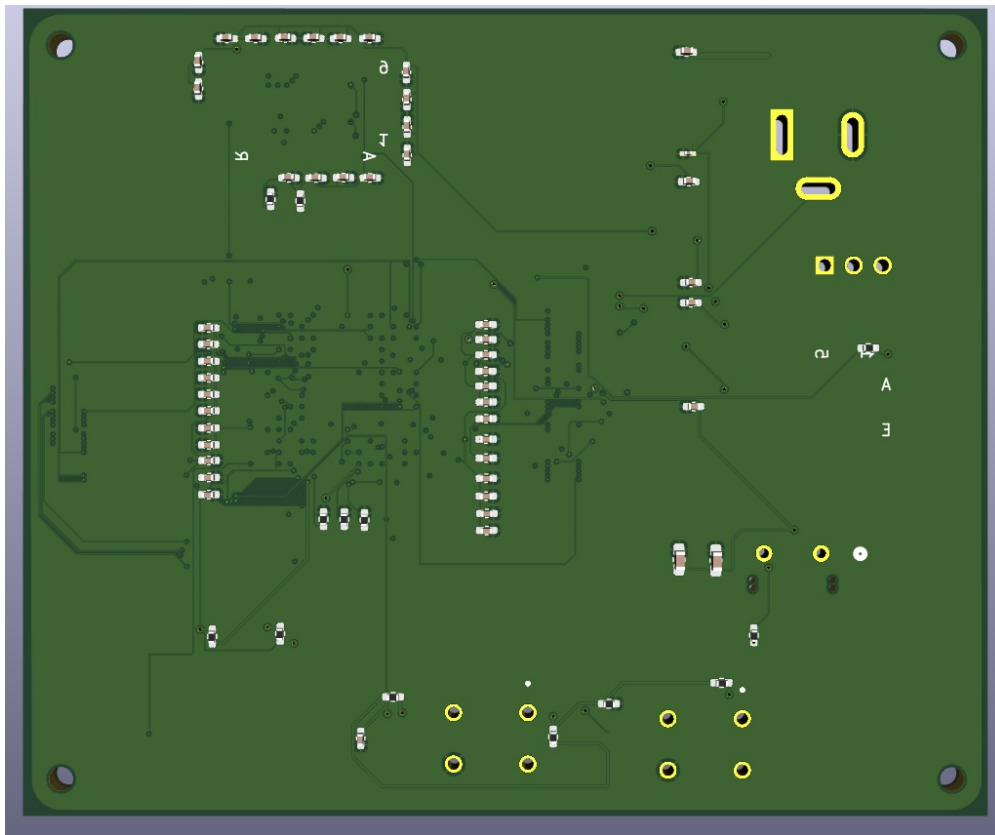


Figure 8.10: 3D View of Final PCB - Back Side

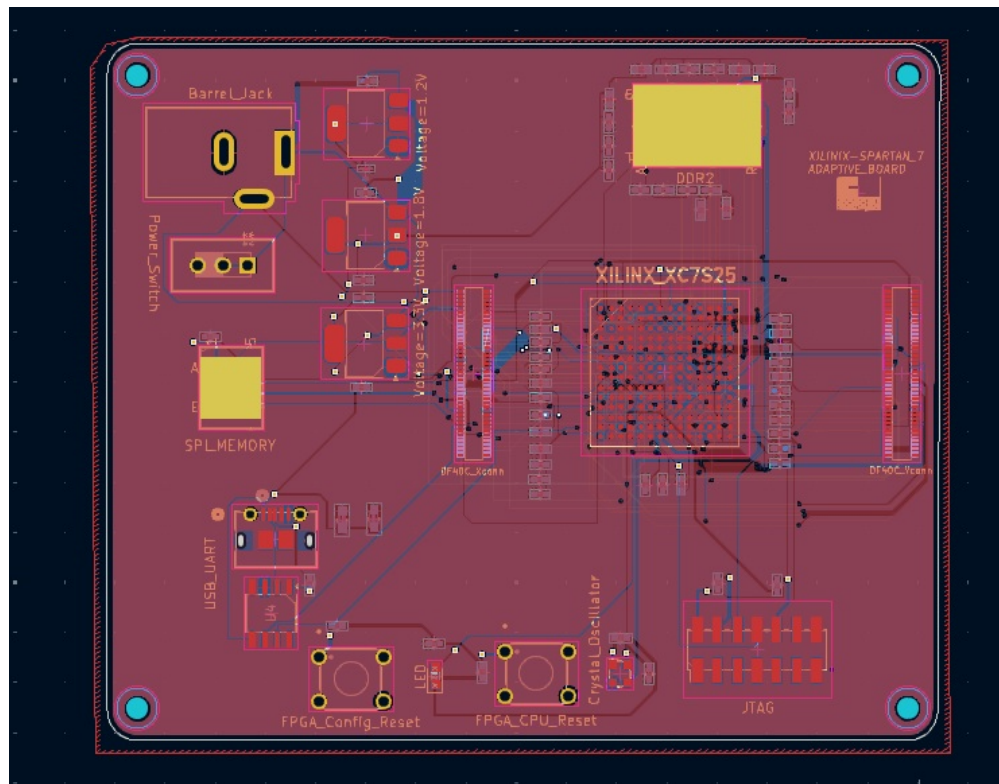


Figure 8.11: Top View of Final PCB Layout

Figure 8.12: Bottom View of Final PCB Layout

Figure 8.13: Isometric View of the Assembled Board

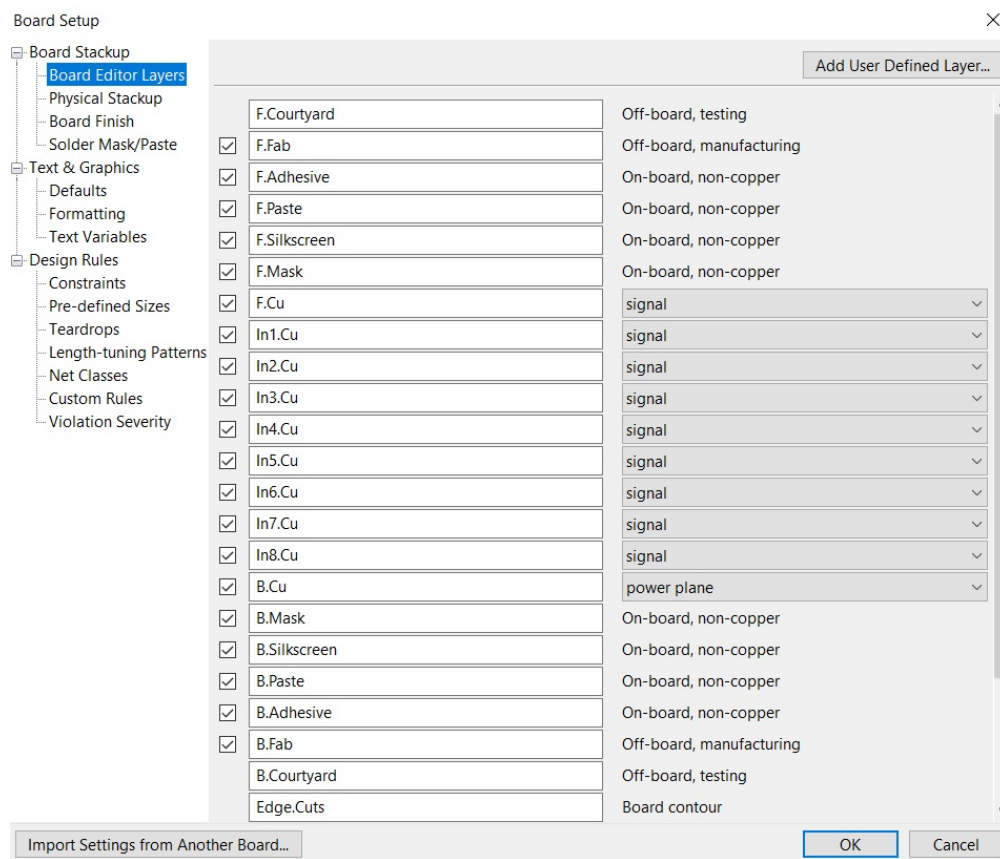


Figure 8.14: Layer Stackup